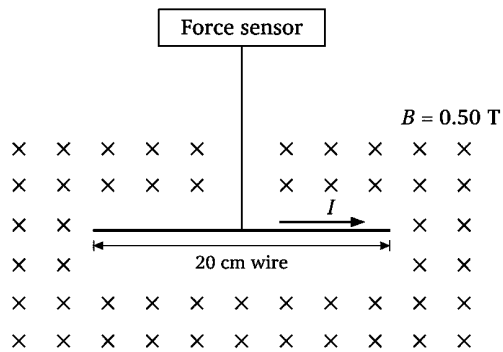


- 1 A point on a guitar string oscillates in simple harmonic motion with a frequency of 50 Hz and an amplitude of 4.0 mm.

What is the total distance travelled by this point during a time interval of 3.0 s?

- A 0.0 m
- B 0.6 m
- C 1.2 m
- D 2.4 m
- E 3.8 m

- 2 A straight, horizontal wire of length 20 cm is suspended from a sensitive force sensor. The wire is in a uniform magnetic field of strength 0.50 T, directed perpendicular to the wire. A total charge of 12 C passes through the wire over a time of 2.0 s. What is the magnitude of the change in the reading on the force sensor?



- A 0.20 N
- B 0.60 N
- C 1.2 N
- D 2.4 N
- E 3.0 N
- F 60 N

- 3 Two objects, X and Y, are in motion. Object X has 3 times the momentum of object Y and 4 times the kinetic energy of object Y.

What is the ratio of their masses, $\frac{m_X}{m_Y}$?

- A $\frac{3}{16}$
- B $\frac{3}{4}$
- C $\frac{4}{9}$
- D $\frac{9}{4}$
- E 12
- F 36

- 4 The intensity I of a physical field varies with distance r from a point source according to a power law relationship $I = kr^n$, where k and n are constants. Experimental measurements yield the following data:

$$\text{At } r = 2.0 \text{ m, } I = 72 \text{ W m}^{-2}$$

$$\text{At } r = 4.0 \text{ m, } I = 18 \text{ W m}^{-2}$$

What is the intensity I at a distance of $r = 3.0 \text{ m}$?

- A 24 W m^{-2}
- B 32 W m^{-2}
- C 36 W m^{-2}
- D 45 W m^{-2}
- E 48 W m^{-2}

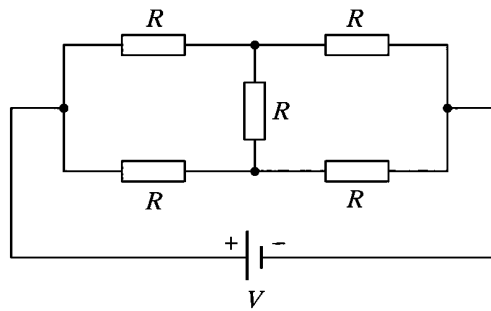
- 5 A fixed mass of gas is sealed in a rigid container and heated. Which statement best explains the increase in pressure?

- A The particles have greater average speed and transfer more momentum to the walls per unit time.
- B The particles become larger, so less empty space remains.
- C The number of particles increases while their average speed remains constant.
- D The particles move more slowly and therefore remain in contact with the walls for longer.
- E The average spacing between particles must increase because the gas is hotter.

- 6 A transmission line receives 20 kW at 2.0 kV. The total resistance of its cables is $5.0\ \Omega$. The voltage at the sending end is increased to 10 kV, while the power entering the line remains 20 kW. The cable resistance is unchanged.

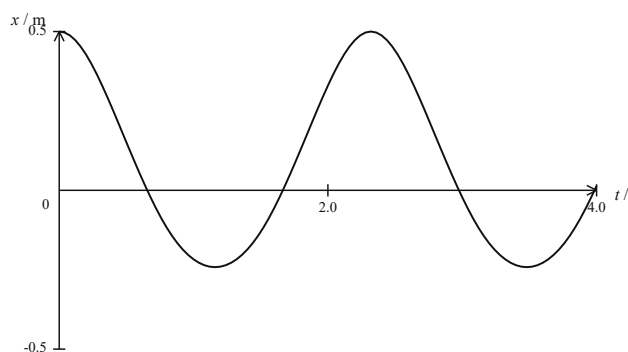
By how much does the power lost in the cables decrease?

- A 20 W
B 100 W
C 400 W
D 480 W
E 500 W
- 7 A circuit is constructed with five identical resistors, each of resistance R , connected to a DC supply of voltage V . The circuit consists of two parallel branches, each containing two resistors in series. A fifth identical resistor connects the midpoint of the first branch to the midpoint of the second branch. The total power dissipated by the circuit is P . If the connecting resistor between the two branches is removed, which expression gives the new total power dissipated?



- A $\frac{1}{2}P$
B $\frac{4}{5}P$
C P
D $\frac{5}{4}P$
E $2P$

- 8 The graph shows the displacement of a particle undergoing simple harmonic motion as a function of time.



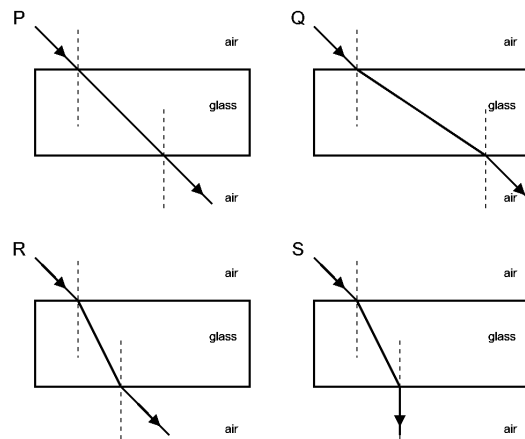
What is the average speed of the particle during the first 4.0 s of its motion?

- A 0 m s^{-1}
 - B 0.25 m s^{-1}
 - C 0.50 m s^{-1}
 - D 1.0 m s^{-1}
 - E $\frac{\pi}{2} \text{ m s}^{-1}$
 - F 2.0 m s^{-1}
- 9 Three devices use different regions of the electromagnetic spectrum:
1. a thermal camera detecting radiation emitted by a warm hand;
 2. a lamp producing ultraviolet radiation;
 3. a scanner producing X-rays.

Which row lists the devices in order of increasing radiation frequency and correctly compares the speeds of their radiation in a vacuum?

- A 1, 2, 3; all three speeds are equal
- B 1, 3, 2; all three speeds are equal
- C 3, 2, 1; all three speeds are equal
- D 1, 2, 3; speed increases in this order
- E 2, 1, 3; speed decreases in this order

- 10** Each diagram is intended to show a ray travelling from air through a rectangular glass block and back into air. Light travels more slowly in glass than in air. The two faces crossed are parallel. Which diagram shows a possible path? Ignore partial reflection.



- A** P
B Q
C R
D S
E None of them
- 11** Straight water waves cross a boundary from deep water into shallow water. In the deep water, the distance from the first crest to the fifth crest is 24 cm. The wave speed there is 30 cm s^{-1} . In the shallow water, the distance from the first crest to the fifth crest is 16 cm. What are the wave frequency and the speed in the shallow water?
- A** 1.25 Hz, 20 cm s^{-1}
B 5.0 Hz, 20 cm s^{-1}
C 5.0 Hz, 45 cm s^{-1}
D 7.5 Hz, 30 cm s^{-1}
E 1.25 Hz, 5.0 cm s^{-1}

- 12** An electric kettle has a power input of 2.5 kW. When used to boil water, it produces steam at a steady rate of 1.0 g s^{-1} .

What is the rate of energy loss from the kettle to the surroundings?

(Specific latent heat of vaporisation of water = $2.0 \times 10^6 \text{ J kg}^{-1}$; specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$)

- A** 0 W
 - B** 500 W
 - C** 1500 W
 - D** 2000 W
 - E** 2500 W
 - F** 4500 W
- 13** A nucleus of Thorium-232 (${}_{90}^{232}\text{Th}$) decays through a series of emissions to become a stable nucleus of Lead-208 (${}_{82}^{208}\text{Pb}$).
The decay series consists entirely of alpha (α) particles and beta-minus (β^-) particles.
What are the total numbers of α and β^- particles emitted during this process?
- A** 4 α , 6 β^-
 - B** 6 α , 4 β^-
 - C** 6 α , 8 β^-
 - D** 6 α , 12 β^-
- 14** A small heating element at the bottom of an open tank of liquid heats a small parcel of the liquid immediately above it.
Just before this heated parcel begins to rise, how do its mass, weight, and the upthrust acting on it compare to their values before it was heated?
- A** Mass: Unchanged, Weight: Unchanged, Upthrust: Increases
 - B** Mass: Unchanged, Weight: Unchanged, Upthrust: Decreases
 - C** Mass: Unchanged, Weight: Decreases, Upthrust: Unchanged
 - D** Mass: Decreases, Weight: Decreases, Upthrust: Unchanged
 - E** Mass: Decreases, Weight: Decreases, Upthrust: Increases
 - F** Mass: Unchanged, Weight: Decreases, Upthrust: Increases

- 15 Two identical copper spheres, **X** and **Y**, are heated to 500 K.

Sphere **X** has a matte black surface.

Sphere **Y** has a polished silver surface.

The initial rate of energy loss, R , is determined for each sphere in a **vacuum** and in an atmosphere of **argon** gas at 298 K.

In the vacuum, it is observed that $R_X > R_Y$.

Assume that the rate of heat transfer due to convection depends only on the dimensions of the sphere and the temperature difference.

Which statement correctly describes the rates in the argon atmosphere compared to the vacuum?

- A** The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ decreases.
- B** The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ stays constant.
- C** The rate R_X increases more than R_Y because black surfaces are better thermal conductors, and the ratio $\frac{R_X}{R_Y}$ increases.
- D** The rate R_Y increases more than R_X because the polished surface reflects gas molecules, and the ratio $\frac{R_X}{R_Y}$ decreases.

- 16 A continuous wave travels from medium X to medium Y. The wave speed in medium Y is $\frac{2}{3}$ of the wave speed in medium X.

What is the fractional change in the wavelength of the wave?

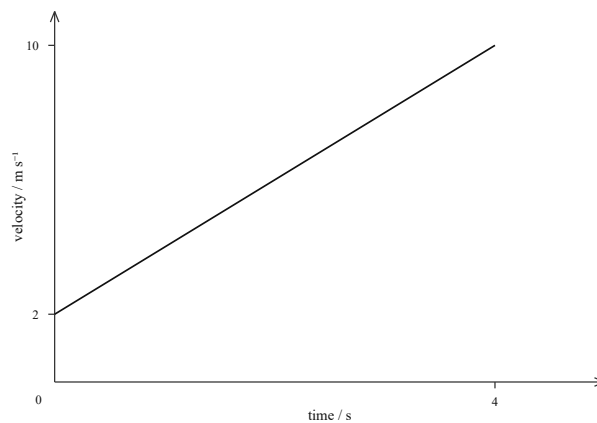
- A** $-\frac{1}{3}$
- B** 0
- C** $\frac{1}{3}$
- D** $\frac{1}{2}$
- E** $\frac{2}{3}$

- 17 The radioactive nuclide ${}_{92}^{235}\text{U}$ decays into the stable nuclide ${}_{82}^{207}\text{Pb}$ via a series of α (alpha) and β^- (beta-minus) emissions.

Which option gives the total number of α and β^- particles emitted during this decay chain?

- A** 5 α , 0 β^-
- B** 7 α , 4 β^-
- C** 7 α , 10 β^-
- D** 8 α , 6 β^-

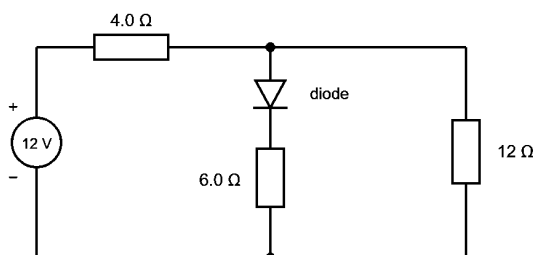
- 18** An electric heater of constant power is used to heat a metal block. The block is initially at room temperature, 20°C .
After 10 minutes, the temperature of the block is 50°C .
The rate of thermal energy loss to the surroundings is proportional to the temperature difference between the block and the surroundings.
Which of the following describes the temperature θ of the block after a total of 20 minutes?
- A** $\theta = 50^{\circ}\text{C}$
B $50^{\circ}\text{C} < \theta < 80^{\circ}\text{C}$
C $\theta = 80^{\circ}\text{C}$
D $80^{\circ}\text{C} < \theta < 100^{\circ}\text{C}$
E $\theta = 100^{\circ}\text{C}$
- 19** An object of mass 2.0 kg moves in a straight line. The velocity-time graph for its motion is shown.



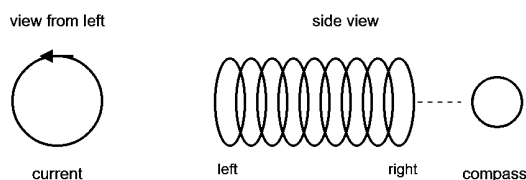
What is the magnitude of the constant net force acting on the object?

- A** 1.0 N
B 2.0 N
C 4.0 N
D 5.0 N
E 12 N
F 20 N

- 20** A 12 V d.c. supply is connected to the circuit shown. The ideal diode initially conducts. A conducting ideal diode has zero resistance; in the reverse direction it blocks all current. The supply connections are reversed. What is the new total power supplied to the circuit?



- A** 6.0 W
B 9.0 W
C 12 W
D 18 W
E 36 W
- 21** The left-hand end of a solenoid is viewed directly. The current around this end flows anticlockwise, as shown. A small compass is placed on the solenoid's axis just beyond its right-hand end. Other magnetic fields are negligible. Which way does the compass's north-seeking end point initially, and after the current is reversed?



- A** right initially; right afterwards
B right initially; left afterwards
C left initially; left afterwards
D left initially; right afterwards
E upwards initially; downwards afterwards

- 22** Twelve identical resistors are connected to form the edges of a cube. A current enters the network at one vertex and leaves at the diagonally opposite vertex.

Let P_{in} be the power dissipated in a single resistor connected directly to the input vertex.

Let P_{mid} be the power dissipated in a single resistor that is connected to neither the input nor the output vertex.

What is the value of the ratio $\frac{P_{\text{in}}}{P_{\text{mid}}}$?

- A** 1
- B** 2
- C** 4
- D** 9
- E** 16

- 23** A solid composite object has total volume $4.0 \times 10^{-3} \text{ m}^3$ and is made from materials X and Y. Their densities are 2000 kg m^{-3} and 800 kg m^{-3} respectively.

It is fully submerged and neutrally buoyant in a fluid of density 1100 kg m^{-3} . For this question, neutral buoyancy means that the object's total mass equals the mass of the displaced fluid:

$$m_{\text{total}} = \rho_{\text{fluid}} V_{\text{total}}.$$

What is the mass of material X?

- A** 1.1 kg
- B** 2.0 kg
- C** 2.4 kg
- D** 4.0 kg
- E** 4.4 kg
- F** 6.0 kg

- 24** An ideal transformer has one primary coil and two separate secondary coils. The primary has 600 turns and is supplied at 120 V a.c.

Secondary X has 60 turns and supplies a 6.0Ω resistor. Secondary Y has 120 turns and supplies a 24Ω resistor. Both loads are connected simultaneously.

What is the primary current?

- A** 0.20 A
- B** 0.30 A
- C** 0.40 A
- D** 0.60 A
- E** 3.0 A

- 25** A fixed mass of an ideal gas initially has pressure P and volume V . The gas is compressed at constant temperature to a final volume of $V/3$. Which expression gives the final pressure of the gas?
- A** $\frac{P}{9}$
B $\frac{P}{3}$
C P
D $3P$
E $9P$
- 26** A short pulse of light travels normally through a transparent block of thickness 12 cm. Its speed in the block is $2.0 \times 10^8 \text{ m s}^{-1}$. The block is replaced by a second block of the same thickness in which the pulse travels at $1.5 \times 10^8 \text{ m s}^{-1}$. The rest of the pulse's path is unchanged. By how much does its arrival at a detector become later?
- A** 0.10 ns
B 0.20 ns
C 0.40 ns
D 0.60 ns
E 0.80 ns
- 27** A DC power supply of voltage V and negligible internal resistance is connected in series with two resistors of resistances R and $2R$. Which expression gives the current flowing through the resistor of resistance R ?
- A** $\frac{V}{3R}$
B $\frac{V}{2R}$
C $\frac{2V}{3R}$
D $\frac{V}{R}$
E $\frac{3V}{2R}$